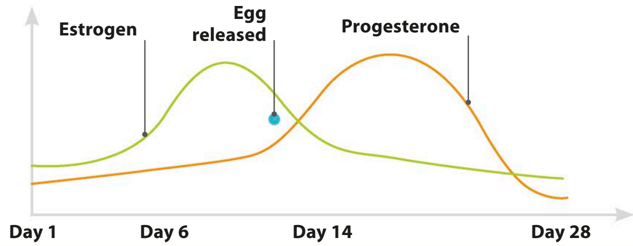
Primary female sex hormones

The two primary female sex hormones are estrogen and progesterone. The changing level of each of these hormones prompts sexual characteristics to develop during puberty and regulates the menstrual cycle.



Estrogen

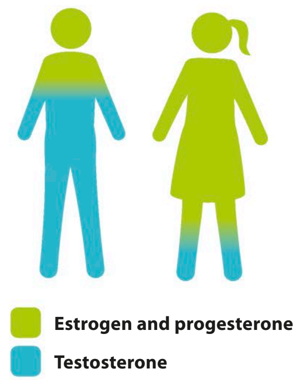
Estrogen is the main female sex hormone at work during puberty. It causes the ovaries to produce eggs and prepare them for the possibility of sexual reproduction. During puberty, estrogen is responsible for promoting the development of sexual characteristics, such as breasts and pubic hair. After puberty, it regulates the menstrual cycle.

Progesterone

Progesterone is present in female and male children at low levels. In females, progesterone comes into effect at the start of the first period. It builds and maintains the lining of the uterus, ready to receive an egg if it's fertilized. If the egg isn't fertilized, progesterone levels drop dramatically, causing the lining of the uterus to be shed during a period.

Shared hormones

Estrogen, progesterone, and testosterone are present in both females and males, but in vastly different amounts. Throughout their lives, females continue to produce low levels of testosterone and males produce some estrogen and progesterone. In females, testosterone is linked to maintaining bone and muscle mass, and contributes to the sex drive. In males, estrogen controls body fat and contributes to the sex drive, while progesterone monitors testosterone production. Hormone levels differ between people and change over a lifetime.



△ **Hormone levels**
Females produce twice as much estrogen and progesterone as males, but ten times less testosterone.

Other hormones

Hormones don't just prompt the start of puberty and the development of sexual characteristics. There are lots of different types at work in everybody, regardless of sex, that control and coordinate many bodily functions to keep the body healthy.

Maintaining a healthy body

- Antidiuretic hormone (ADH) keeps the body's water levels balanced.
- Melatonin allows the body to sleep at night and stay awake during the day.
- Thyroxine determines how quickly or slowly the body metabolizes food.

Managing food processes

- Leptin regulates the appetite by making the body feel full after eating.
- Gastrin triggers gastric acid in the stomach, which breaks down food.
- Insulin and glucagon control how much sugar is released into the blood after eating.

Coping with stress

- Adrenaline raises the heart rate and produces energy when a person is under stress.
- Cortisol manages the brain's use of sugars, providing more energy.
- Oxytocin enables bonds with other people by reducing fear and creating feelings of trust.